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Academic Profile

PhD researcher in Industrial Design at Eindhoven University of Technology and visiting PhD researcher at EPFL, working at the intersection of data centric AI, computer vision, world models, reinforcement learning, and human centred autonomous systems. At EPFL, I am a funded PhD researcher on the Swiss AI Initiative Large Project *Reality Grounded World Model*. My research combines globally diverse traffic data, generative and multimodal models, simulation, and behavioural experiments to build autonomous systems that generalise across environments and remain interpretable to people. I have led first author work on large scale datasets and perception pipelines, published across artificial intelligence, autonomous systems, and human factors, contributed to master's level teaching, and supervised or co-supervised seven MSc students and project students, one BSc student, and three research interns. I am applying for Assistant Professor positions in artificial intelligence, machine learning, computer vision, autonomous systems, and human centred computing.

Research Interests

World models; computer vision; data centric AI; generative and multimodal AI; vision language models; reinforcement learning and control; autonomous systems; human AI interaction; road user behaviour modelling; synthetic data; domain translation; simulation based evaluation; traffic safety and mobility.

Research Programme

Reality grounded and data centric AI. I develop globally diverse datasets, annotations, and world models for urban environments. This programme includes CROWD, with 75,147 continuous driving segments and 26,273.75 hours of video from 8,049 inhabited places across 238 countries and territories, and TraffCOCO, an ongoing traffic focused dataset comprising approximately 13 million images. At EPFL, through the Swiss AI Initiative Large Project *Reality Grounded World Model*, I am developing reality grounded world models for controllable 4D urban simulation using visual observations, maps, geometry, trajectories, and dynamic road user behaviour.

Human centred autonomous systems. I study how autonomous systems perceive, predict, and communicate with people. My work includes global pedestrian behaviour analysis across 2,495 cities in 233 countries and territories, evaluation of 13 vision language models against responses from 1,438 participants, virtual reality studies of pedestrian interaction, and multimodal interfaces for automated vehicles.

Learning based control and simulation. I develop reinforcement learning methods for autonomous decision making and control. My work spans DQN, DDPG, PPO, behavioural cloning, self play, and simulation based evaluation, including validation of learned vessel controllers on a 1:75.5 scale physical platform.

Academic and Research Appointments

Jan – Jun 2026

- **Visiting PhD Researcher, World Models and Computer Vision**, École Polytechnique Fédérale de Lausanne (EPFL), Lausanne, Switzerland.
Selected as a funded PhD researcher on the Swiss AI Initiative Large Project *Reality Grounded World Model*, led by EPFL.
Developing a reality grounded world model for controllable 4D urban simulation using street level imagery, maps, multi view observations, geometry, navigation trajectories, and dynamic road user behaviour.
Building data and evaluation pipelines for temporal consistency, structural faithfulness, geographical generalisation, long horizon memory, and safety critical scenario generation.
Investigating how globally diverse urban traffic data can support autonomous driving, robotics, embodied AI, and interactive simulation.

2023 – Present

- **PhD Researcher, Machine Learning, Computer Vision, and Human Factors**, Eindhoven University of Technology (TU/e), Eindhoven, the Netherlands.
Created CROWD, a global traffic video dataset containing 75,147 continuous segments, 26,273.75 hours of footage, and 58,467 unique videos from 8,049 inhabited places across 238 countries and territories.
Developing TraffCOCO, a traffic focused computer vision dataset comprising approximately 13 million images, with COCO compatible annotations for road users, detailed traffic signs, traffic cones, zebra crossings, road markings, and other infrastructure.
Analysed pedestrian crossing behaviour across 2,495 cities in 233 countries and territories, measuring an average crossing initiation time of 3.18 seconds and an average crossing speed of 1.20 m/s.
Developed generative traffic video pipelines that improved temporal consistency by up to 7.97% and structural fidelity by up to 153.49% over selected baselines.
Benchmarked 13 vision language models across 227 automated vehicle interface images and 15 trials against responses from 1,438 participants, obtaining correlations of up to 0.85.

2021 – 2023

- **Junior Research Fellow, Reinforcement Learning and Autonomous Systems**, Centre for Industrial Consultancy and Sponsored Research, Indian Institute of Technology Madras, Chennai, India.
Developed DQN, DDPG, and PPO controllers for an underactuated autonomous surface vessel using nonlinear three degree of freedom KCS ship dynamics.
Designed reinforcement learning environments, observation spaces, reward functions, and evaluation procedures for path following, wind disturbance handling, and obstacle avoidance.
Benchmarked learned policies against PD and ILOS guidance, achieving 0.2969L RMSE on ellipse tracking and 0.4557L RMSE on figure eight tracking, with lower simulation error than the traditional baseline.
Validated learned controllers in simulation and on a 1:75.5 scale physical vessel using IMU and GPS sensing.

Education

- 2023 – 2026 (expected)  **PhD, Industrial Design**, Eindhoven University of Technology, Eindhoven, the Netherlands.
Thesis: *Connected data from autonomous vehicles and road users to enhance safety*.
Research focus: machine learning, computer vision, data centric AI, human factors, and autonomous mobility.
- 2021 – 2023  **MSc by Research, Robotics and Artificial Intelligence**, Indian Institute of Technology Madras, Chennai, India.
Thesis: *Data driven control for marine vehicle maneuvering*.
- 2016 – 2020  **Bachelor of Technology, Mechanical Engineering**, Jamia Millia Islamia, Delhi, India.
Thesis: *Design of product prototype for inventory management 4.0*.

Research Publications

Peer Reviewed Journal Articles

- 1 **M. S. Alam**, P. Singh, and P. Bazilinskyy, “A survey of day-night illumination domain translation for outdoor vision: Methods, datasets, and evaluation protocols,” *Machine Vision and Applications*, vol. 37, p. 97, Jun. 17, 2026, Peer reviewed and published open access on 17 June 2026; first and corresponding author.  DOI: 10.1007/s00138-026-01843-8.
- 2 **M. S. Alam** and I. Carlucho, “Harnessing traditional controllers for fast track training of deep reinforcement learning control strategies,” *Journal of Marine Engineering and Technology*, pp. 1–12, 2024, First author.  DOI: 10.1080/20464177.2024.2367276.
- 3 R. Deraj, R. S. S. Kumar, **M. S. Alam**, and A. Somayajula, “Deep reinforcement learning based controller for ship navigation,” *Ocean Engineering*, vol. 273, p. 113 937, 2023.  DOI: 10.1016/j.oceaneng.2023.113937.

Peer Reviewed Conference Proceedings

- 1 **M. S. Alam** and P. Bazilinskyy, “Cross or nah? llms get in the mindset of a pedestrian in front of an automated car with an ehmi,” in *Adjunct Proceedings of the 17th International Conference on Automotive User Interfaces and Interactive Vehicular Applications*, ser. AutomotiveUI 2025 Adjunct, First and corresponding author, New York, NY, USA: Association for Computing Machinery, 2025, pp. 19–34.  DOI: 10.1145/3744335.3758477.
- 2 **M. S. Alam**, M. H. Martens, and P. Bazilinskyy, “Generating realistic traffic scenarios: A deep learning approach using generative adversarial networks,” in *Human Interaction and Emerging Technologies: Artificial Intelligence and Future Applications*, First author, vol. 161, AHFE International, 2025, pp. 349–358.  DOI: 10.54941/ahfe1005927.
- 3 **M. S. Alam**, M. H. Martens, and P. Bazilinskyy, “Pedestrian planet: What youtube driving from 233 countries and territories teaches us about the world,” in *Proceedings of the 17th International Conference on Automotive User Interfaces and Interactive Vehicular Applications*, ser. AutomotiveUI 2025, First and corresponding author; Honourable Mention, Brisbane, QLD, Australia: Association for Computing Machinery, 2025.  DOI: 10.1145/3744333.3747827.
- 4 **M. S. Alam**, S. H. Parmar, M. H. Martens, and P. Bazilinskyy, “Deep learning approach for realistic traffic video changes across lighting and weather conditions,” in *2025 8th International Conference on Information and Computer Technologies*, First and corresponding author, Hilo, HI, USA: IEEE, 2025, pp. 180–185.  DOI: 10.1109/ICICT64582.2025.00034.

- 5 P. Bazilinskyy, **M. S. Alam**, and R. Merino-Martinez, "Pedestrian crossing behaviour in front of electric vehicles emitting synthetic sounds: A virtual reality experiment," in *Proceedings of the 54th International Congress and Exposition on Noise Control Engineering*, Sao Paulo, Brazil: INTER-NOISE, 2025.  DOI: 10.3397/IN_2025_1076086.
- 6 P. Bazilinskyy, **M. S. Alam**, and R. Merino-Martinez, "Psychoacoustic assessment of synthetic sounds for electric vehicles in a virtual reality experiment," in *Proceedings of the 11th Convention of the European Acoustics Association*, Malaga, Spain: European Acoustics Association, 2025.  DOI: 10.61782/fa.2025.0710.
- 7 M. Fortes-Ferreira, **M. S. Alam**, and P. Bazilinskyy, "Vibe coding in practice: Building a driving simulator without expert programming skills," in *Adjunct Proceedings of the 17th International Conference on Automotive User Interfaces and Interactive Vehicular Applications*, ser. AutomotiveUI 2025 Adjunct, New York, NY, USA: Association for Computing Machinery, 2025, pp. 60–66.  DOI: 10.1145/3744335.3758482.
- 8 **M. S. Alam**, T. Subramanian, M. Martens, W. Remlinger, and P. Bazilinskyy, "From a to b with ease: User centric interfaces for shuttle buses," in *Adjunct Proceedings of the 16th International Conference on Automotive User Interfaces and Interactive Vehicular Applications*, ser. AutomotiveUI 2024 Adjunct, First and corresponding author, New York, NY, USA: Association for Computing Machinery, 2024, pp. 111–116.  DOI: 10.1145/3641308.3685033.
- 9 **M. S. Alam**, S. K. Ramkumar Sudha, and A. Somayajula, "Ai on the water: Applying drl to autonomous vessel navigation," in *Proceedings of the Sixth International Conference in Ocean Engineering*, First author, ICOE, 2023.  DOI: 10.48550/arXiv.2310.14938. arXiv: 2310.14938.
- 10 J. Jose, **M. S. Alam**, and A. S. Somayajula, "Navigating the ocean with drl: Path following for marine vessels," in *Proceedings of the Sixth International Conference in Ocean Engineering*, ICOE, 2023.  DOI: 10.48550/arXiv.2310.14932. arXiv: 2310.14932.
- 11 S. K. Ramkumar Sudha, **M. S. Alam**, B. Reddy, and A. S. Somayajula, "Comparison of path following in ships using modern and traditional controllers," in *Proceedings of the Sixth International Conference in Ocean Engineering*, ICOE, 2023. arXiv: 2310.14940.  URL: <https://arxiv.org/abs/2310.14940>.

Preprints and Manuscripts

- 1 **M. S. Alam**, O. Bazilinska, and P. Bazilinskyy, "A global dataset of continuous urban dashcam driving," Preprint; first and corresponding author, 2026. arXiv: 2604.01044.  URL: <https://arxiv.org/abs/2604.01044>.
- 2 **M. S. Alam** and P. Bazilinskyy, "Nineteen years of asmr on youtube: A multilingual, theme level analysis of 72,070 videos," Manuscript under review; first and corresponding author, 2026.
- 3 **M. S. Alam** and P. Bazilinskyy, "Randomised versus fixed trial order in repeated exposure vr studies of pedestrian interaction with automated vehicles," Manuscript under review; first and corresponding author, 2026.
- 4 **M. S. Alam** and P. Bazilinskyy, "What can public traffic cameras reveal? a short horizon privacy audit using open source vehicle tracking," Manuscript under review; first and corresponding author, 2026.
- 5 **M. S. Alam**, L. Zhang, J. Li, F. Dou, and P. Bazilinskyy, "Collision patterns and reporting blind spots in 971 california autonomous vehicle crash reports," Manuscript under review; first author, 2026.
- 6 W. Verstelle, **M. S. Alam**, and P. Bazilinskyy, "Personalised electric vehicle acoustics with generative ai: Dynamic sonification and user acceptance study," Manuscript under review, 2026.
- 7 X. Zeng, **M. S. Alam**, and P. Bazilinskyy, "Enhancing driver experience in sae level 3 automated vehicles through multimodal and emotion aware in vehicle agents," Manuscript under review, 2026.

- 8 **M. S. Alam**, V. Deogaonkar, and A. Somayajula, "Enhancing autonomy: Comparative analysis of machine learning based controllers in extended action states," Manuscript under review; first author, 2025.
- 9 **M. S. Alam**, D. Dey, M. H. Martens, and P. Bazilinskyy, "You'll never walk alone: Inter pedestrian distance, ehmis, and crossing decisions in virtual reality," Manuscript under review; first and corresponding author, 2025.
- 10 **M. S. Alam**, M. H. Martens, O. Bazilinska, and P. Bazilinskyy, "Understanding global pedestrian behaviour in 565 cities with dashcam videos on youtube," Manuscript under review; first and corresponding author, 2025.
- 11 **M. S. Alam**, Z. Wang, L. Zhang, and P. Bazilinskyy, "Exploring veo 3's capabilities for generating urban traffic scenes in 76 cities worldwide," Preprint; first and corresponding author, 2025.  URL: <https://www.researchgate.net/publication/393086985>.

Research Funding, Grants, Awards, and Recognition

Competitive Research Funding and Compute Awards

2026: **Funded PhD Researcher, Swiss AI Initiative Large Project: Reality Grounded World Model.** EPFL led consortium project developing reality grounded world models for controllable 4D urban simulation; contribution during the EPFL research appointment focuses on geographically diverse traffic data, human centred dynamics, and reality centric evaluation.

2026: **Recipient, FAIR Data Fund (5th edition)**, 4TU.ResearchData, supporting FAIRification, documentation, quality assurance, and reproducible release of the TraffCOCO traffic dataset.

2026: **Recipient, NWO/SURF Small Computing Time allocation**, providing access to Dutch national computing facilities for large scale machine learning research.

Awards and Recognition

2025: **Honourable Mention, AutomotiveUI '25**, for the first author paper *Pedestrian Planet: What YouTube Driving from 233 Countries and Territories Teaches Us About the World*.

2022: **Winner, Virtual RobotX Competition**, organised by the Office of Naval Research, the Naval Postgraduate School, and Open Robotics.

2022: **Winner, Data Analysis Hackathon**, IEEE OCEANS 2022.

Teaching Experience

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| 2024 – 2025 |  Embodying Intelligent Behavior in Social Context (DBM140) , Eindhoven University of Technology, Eindhoven, the Netherlands.
Contributed to master's level course delivery, project based discussions, and student learning on social robotics, intelligent behaviour, artificial intelligence, and social context. |
| 2024 |  Automotive Human Factors (oHM310) , Eindhoven University of Technology, Eindhoven, the Netherlands.
Taught in a master's level course examining interactions among vehicles, drivers, road users, and dynamic traffic environments, including technological, social, ethical, and legal dimensions of automated mobility. |
| |  Core Courses, MSc Industrial Design , Eindhoven University of Technology, Eindhoven, the Netherlands.
Contributed teaching across the graduate core course portfolio and supported project based learning in the MSc Industrial Design programme. |

Student Supervision

Supervised or co-supervised 11 students at Eindhoven University of Technology: seven MSc students and project students, one BSc student, and three research interns. Projects span human centred AI, multimodal interaction, generative AI, computer vision, automated mobility, and scientific software development.

MSc Students and Project Groups

2025 – Present: **Christian Janssen**, dynamic external human machine interfaces for pedestrians; co-supervision with the IE&IS department.

2024: **Matin Maskitou, Yani Wang, Xander Fanchamps, and Giovanni Sapienza**, evaluation of external human machine interfaces using LLM based personas and human participants recruited through crowdsourcing; completed MSc project group.

2024: **Xingjian Zeng**, multimodal and emotion aware in-vehicle agents for SAE Level 3 automated driving; completed final master's project and resulting manuscript.

2024: **Alexandra van Dijk**, generative AI for persona based traffic research; completed MSc project.

BSc Students

2026: **Wessel Verstelle**, personalised electric vehicle acoustics using generative AI, dynamic sonification, and user acceptance evaluation; completed BSc project and resulting manuscript.

Research Internships

2026: **Priyanshu Singh**, day-night illumination domain translation for outdoor vision; completed internship and resulting peer reviewed open access journal publication in *Machine Vision and Applications*.

2026: **Brileena Benny**, research internship; project in progress.

2025: **Margarida Fortes Ferreira**, LLM assisted development of a 3D driving simulator; completed internship and resulting peer reviewed AutomotiveUI Adjunct publication.

Full supervision record: <https://shaadalam9.github.io/education/>

Teaching Interests

Machine learning; computer vision; deep learning; generative and multimodal AI; reinforcement learning; robotics and autonomous systems; human centred AI; data science; research methods; experimental design; responsible AI.

Research Visits and Collaborations

2026 – Present



University College London, London, United Kingdom.

Collaborating with **Dr. Mark Colley** on extending *OptiCarVis*, a human in the loop framework for optimising explainable automated vehicle visualisations. The collaboration focuses on improving the visualisation design space, optimisation workflow, user interaction, and evaluation of trust, safety, predictability, cognitive load, and acceptance.

2025 – Present



Uppsala University, Uppsala, Sweden.

Collaborating with **Dr. Alexandros Rouchitsas** on *Cycle Following*, a human factors study investigating cyclist following behaviour, perceived safety, interaction dynamics, and behavioural responses in conventional and automated mobility environments.



Indian Institute of Technology Indore, Indore, India.

Collaborating with **Dr. Pushpa Choudhary** on a cross cultural study of external human machine interfaces, examining how message language and wording influence pedestrian comprehension, perceived safety, trust, and crossing decisions.

May 2024



Leipzig University, Leipzig, Germany.

Research visit with **Dr. Patrick Ebel** focused on computational interaction, human behaviour modelling, and mobility oriented machine learning for human centred transport systems.

Research Visits and Collaborations (continued)

- Mar – Apr 2024  **University of Stuttgart**, Stuttgart, Germany.
Collaborated with **Prof. Dr. Wolfram Remlinger** on human centred mobility, vehicle interior design, and user focused interaction in future transport systems.

Academic Service and Leadership

Peer Review

2024 – Present: Reviewer for NeurIPS; International Journal of Control, Automation, and Systems; Journal of Intelligent & Robotic Systems; Ocean Engineering; Engineering Applications of Artificial Intelligence; and Transportation Research Interdisciplinary Perspectives. Reviewed more than 20 journal and conference submissions across artificial intelligence, control, robotics, and transportation.

Conference and Professional Service

2024 – 2025: Web Chair, IEEE RO-MAN 2025 conference.

2022 – 2023: Placement Coordinator, IIT Madras, supporting master's and PhD student placement activities.

Industry Experience

- Mar – Apr 2025  **Machine Learning Engineer, Reinforcement Learning**, CESARE, Paris, France.
Integrated PPO based reinforcement learning into the MASA SWORD agent based simulation platform for autonomous decision making in scenario driven environments.
Implemented two sided self play, seed controlled evaluation, curriculum based tasks, Docker execution, configuration driven experiments, and automated reporting.
- May – Jul 2022  **Machine Learning Engineer**, GoalWit Technologies, Delhi, India.
Developed subscription conversion and plan tier prediction models, achieving 93.8% accuracy on multi class plan tier prediction.
Built SQL based feature extraction, batch scoring, and customer segmentation workflows that converted model predictions into ranked sales leads.

Professional Development

- Jun 2024  **10th International Summer School on AI and Big Data**, ScaDS.AI Dresden/Leipzig, Leipzig, Germany.
Training in generative AI, responsible AI, mathematical foundations of AI, and applied AI and big data methods.

Technical Expertise

Programming and data: Python, C++, C, SQL, R, NumPy, Pandas, data pipelines, dataset curation, COCO annotation formats. **Machine learning and AI:** PyTorch, TensorFlow, JAX, world models, computer vision, reinforcement learning, transformers, vision language models, large language models, generative adversarial networks, behavioural cloning, and Bayesian optimisation. **Computer vision and video:** object detection, semantic segmentation, multi object tracking, YOLO, BoT SORT, DeepSORT, OpenCV, generative video, domain translation, and synthetic data. **Research infrastructure:** MLflow, DVC, Docker, Kubernetes, CI/CD for ML, SLURM, and reproducible experimentation. **Simulation and human centred research:** Unity, Unreal Engine, ROS, sensor integration, virtual reality experiments, user studies, statistical analysis, and experimental design.